

Задание 18 - Корневая файловая система

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Задание 1

- 1) Создаем файл init.c

```
komal@komal-VirtualBox:~/Task_18$ ls
vexpress-v2p-ca9.dtb  zImage
komal@komal-VirtualBox:~/Task_18$ touch init.c
komal@komal-VirtualBox:~/Task_18$ ls
init.c  vexpress-v2p-ca9.dtb  zImage
komal@komal-VirtualBox:~/Task_18$ sudo nano init.c
[sudo: authenticate] Password:
komal@komal-VirtualBox:~/Task_18$
```

```
GNU nano 8.7.1                               init.c *
#include <stdio.h>
#include <unistd.h>

int main(void){
    printf("HEEEEELLLLLOOOO!!!! DIMA KOVALENKO!!!!\n");
    sleep(20);
}
```

- 2) Компилируем программу с -static

```
[sudo: authenticate] Password:
komal@komal-VirtualBox:~/Task_18$ arm-linux-gnueabihf-gcc -static init.c -o init
```

- 3) Создаем образ корневой файловой системы

```
komal@komal-VirtualBox:~/Task_18$ ls
init  init.c  vexpress-v2p-ca9.dtb  zImage
komal@komal-VirtualBox:~/Task_18$ echo init | cpio -o -H newc | gzip > initramfs.cpio.gz
```

- 4) И запускаем нашу прошивку до того момента пока не кончится время в init.c

```

komal@komal-VirtualBox:~/Task_18$ ls
init init.c initramfs.cpio.gz vexpress-v2p-ca9.dtb zImage
komal@komal-VirtualBox:~/Task_18$ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0 -nographic"

```

```

[ 3.708584] rtc-pl031 10017000.rtc: registered as rtc0
[ 3.709094] rtc-pl031 10017000.rtc: setting system clock to 2026-06-14T21:06:55 UTC (1781471215)
[ 3.710714] i2c_dev: i2c /dev entries driver
[ 3.724014] mmci-pl18x 10005000.mmci: Got CD GPIO
[ 3.724232] mmci-pl18x 10005000.mmci: Got WP GPIO
[ 3.726524] mmci-pl18x 10005000.mmci: mmc0: PL181 manf 41 rev0 at 0x10005000 irq 31,32 (pio)
[ 3.734608] usbcore: registered new interface driver usbhid
[ 3.734652] usbhid: USB HID core driver
[ 3.739991] hw perfevents: enabled with armv7_cortex_a9 PMU driver, 7 (8000003f) counters available
[ 3.742408] NET: Registered PF_INET6 protocol family
[ 3.747637] Segment Routing with IPv6
[ 3.747782] In-situ OAM (IOAM) with IPv6
[ 3.748125] NET: Registered PF_PACKET protocol family
[ 3.748162] can: controller area network core
[ 3.748487] NET: Registered PF_CAN protocol family
[ 3.748546] can: raw protocol
[ 3.748668] can: broadcast manager protocol
[ 3.748773] can: netlink gateway - max_hops=1
[ 3.749565] Key type dns_resolver registered
[ 3.749896] ThumbEE CPU extension supported.
[ 3.749954] Registering SWP/SWPB emulation handler
[ 3.758608] Loading compiled-in X.509 certificates
[ 3.936183] input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10006000.kmi/serio0/input/input0
[ 3.940109] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
[ 3.943342] printk: console [ttyAMA0] enabled
[ 3.948908] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
[ 3.949867] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
[ 3.950812] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[ 3.957125] clk: Disabling unused clocks
[ 3.957345] PM: genpd: Disabling unused power domains
[ 3.985706] Freeing unused kernel image (initmem) memory: 2048K
[ 3.988871] Run /init as init process
HEEEELLLLL0000!!!! DIMA KOVALENKO!!!!
[ 5.269779] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/serio1/input/input2
s

```

```

[ 3.748162] can: controller area network core
[ 3.748487] NET: Registered PF_CAN protocol family
[ 3.748546] can: raw protocol
[ 3.748668] can: broadcast manager protocol
[ 3.748773] can: netlink gateway - max_hops=1
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[ 3.758608] Loading compiled-in X.509 certificates
[ 3.936183] input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10006000.kmi/serio0/input/input0
[ 3.940109] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
[ 3.943342] printk: console [ttyAMA0] enabled
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[ 3.950812] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[ 3.957125] clk: Disabling unused clocks
[ 3.957345] PM: genpd: Disabling unused power domains
[ 3.985706] Freeing unused kernel image (initmem) memory: 2048K
[ 3.988871] Run /init as init process
HEEEELLLLLLOOOO!!!! DIMA KOVALENKO!!!!
[ 5.269779] input: ImExPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/serio1/input/input2
[ 15.696475] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 15.696829] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 15.696864] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 15.696885] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
[ 24.680540] Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000
[ 24.681308] CPU: 0 UID: 0 PID: 1 Comm: init Not tainted 7.1.0-rc7-ge21ee273e6fa #1 VOLUNTARY
[ 24.681481] Hardware name: ARM-Versatile Express
[ 24.681607] Call trace:
[ 24.681927] unwind_backtrace from show_stack+0x10/0x14
[ 24.682389] show_stack from dump_stack_lvl+0x54/0x68
[ 24.682461] dump_stack_lvl from vpanic+0x234/0x450
[ 24.682522] vpanic from do_panic_on_target_cpu+0x0/0x14
[ 24.683170] ---[ end Kernel panic - not syncing: Attempted to kill init! exitcode=0x00000000 ]---

```

Задание 2

- 1) Делаем default config в busybox

```

komal@komal-VirtualBox:~/busybox-1.36.1$ ARCH=arm make defconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/basic/split-include
HOSTCC scripts/basic/docproc
GEN include/applets.h
GEN include/usage.h
GEN e2fsprogs/Kbuild
GEN e2fsprogs/Config.in
GEN loginutils/Kbuild
GEN loginutils/Config.in
GEN runit/Kbuild
GEN runit/Config.in
GEN findutils/Kbuild
GEN findutils/Config.in
GEN libbb/Kbuild
GEN libbb/Config.in
GEN console-tools/Kbuild
GEN console-tools/Config.in
GEN libpwdgrp/Kbuild
GEN miscutils/Kbuild
GEN miscutils/Config.in
GEN printutils/Kbuild

```


2) Запускаем ARCH=arm make menuconfig для настройки

```
↑(-)
[*] Support wtmp file
[*] Support writing pidfiles
(/var/run) Directory for pidfiles
[*] Include busybox applet
[*] Support --show SCRIPT
[*] Support --install [-s] to install applet links at runtime
[ ] Don't use /usr
[*] Drop SUID state for most applets
[*] Enable SUID configuration via /etc/busybox.conf
[*] Suppress warning message if /etc/busybox.conf is not readable
[ ] exec prefers applets
(/proc/self/exe) Path to busybox executable
[ ] Support NSA Security Enhanced Linux
[ ] Clean up all memory before exiting (usually not needed)
[*] Support LOG_INFO level syslog messages
--- Build Options
[*] Build static binary (no shared libs)
[ ] Force NOMMU build
() Cross compiler prefix
() Path to sysroot
() Additional CFLAGS
() Additional LDFLAGS
() Additional LDLIBS
↓(+)
```

Cross compiler prefix
Please enter a string value. Use the <TAB> key to move from the input field to the buttons below it.

arm-linux-gnueabihf-

< Ok >

< Help >

3) Создаем корневую файловую систему

```

komal@komal-VirtualBox:~/busybox-1.36.1$
komal@komal-VirtualBox:~/busybox-1.36.1$ make
  SPLIT    include/autoconf.h -> include/config/*
  GEN      include/bbconfigopts.h
  GEN      include/common_bufsiz.h
  GEN      include/embedded_scripts.h
  HOSTCC   applets/usage
applets/usage.c: In function 'main':
applets/usage.c:52:17: warning: ignoring return value of 'write' declared
 52 |                 write(STDOUT_FILENO, usage_array[i].usage, strlen(usage_array[i].usage))
    |                 ^~~~~~
  GEN      include/usage_compressed.h
  HOSTCC   applets/applet_tables
  GEN      include/applet_tables.h include/NUM_APPLETS.h
  GEN      include/applet_tables.h include/NUM_APPLETS.h
  CC       applets/applets.o

```

```

komal@komal-VirtualBox:~/busybox-1.36.1$ make install
./_install/bin/arch -> busybox
./_install/bin/ash -> busybox
./_install/bin/base32 -> busybox
./_install/bin/base64 -> busybox
./_install/bin/cat -> busybox
./_install/bin/chattr -> busybox
./_install/bin/chgrp -> busybox
./_install/bin/chmod -> busybox
./_install/bin/chown -> busybox
./_install/bin/conspy -> busybox
./_install/bin/cp -> busybox
./_install/bin/cpio -> busybox
./_install/bin/cttyback -> busybox

```

```

komal@komal-VirtualBox:~/busybox-1.36.1/_install$ find . | cpio -o -H newc | gzip > initramfs.cpio.gz
cpio: File ./initramfs.cpio.gz grew, 786432 new bytes not copied
5245 blocks
komal@komal-VirtualBox:~/busybox-1.36.1/_install$

```

- 4) Копирую его в свою папку, после того как удалим старую корневую файловую систему

```

komal@komal-VirtualBox:~/busybox-1.36.1/_install$ cp initramfs.cpio.gz ~/Task_18/initramfs.cpio.gz
komal@komal-VirtualBox:~/busybox-1.36.1/_install$

```

- 5) И запускаем систему

```

komal@komal-VirtualBox:~/Task_18$ ls
init init.c initramfs.cpio.gz vexpress-v2p-ca9.dtb zImage
komal@komal-VirtualBox:~/Task_18$ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0 rdinit=/bin/ash" -nographic

```

```
[ 1.527829] Key type dns_resolver registered
[ 1.528278] ThumbEE CPU extension supported.
[ 1.528343] Registering SWP/SWPB emulation handler
[ 1.543413] Loading compiled-in X.509 certificates
[ 1.559841] 10009000.serial: ttyAMA0 at MMIO 0x10009000 (irq = 37, base_baud = 0) is a PL011 rev1
[ 1.561620] printk: console [ttyAMA0] enabled
[ 1.602677] input: AT Raw Set 2 keyboard as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10006000.kmi/serio0/input/input0
[ 1.604783] 1000a000.serial: ttyAMA1 at MMIO 0x1000a000 (irq = 38, base_baud = 0) is a PL011 rev1
[ 1.606686] 1000b000.serial: ttyAMA2 at MMIO 0x1000b000 (irq = 39, base_baud = 0) is a PL011 rev1
[ 1.609111] 1000c000.serial: ttyAMA3 at MMIO 0x1000c000 (irq = 40, base_baud = 0) is a PL011 rev1
[ 1.629445] clk: Disabling unused clocks
[ 1.629693] PM: genpd: Disabling unused power domains
[ 1.660732] Freeing unused kernel image (initmem) memory: 2048K
[ 1.665408] Run /bin/ash as init process
/bin/ash: can't access tty; job control turned off
~ # [ 2.014379] input: ImEXPS/2 Generic Explorer Mouse as /devices/platform/bus@40000000/bus@40000000:motherboard-bus@40000000:iofpga@7,00000000/10007000.kmi/serio1/input/input2

~ #
~ # [ 11.995694] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 11.996563] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 11.996631] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 11.996676] amba 100e5000.watchdog: deferred probe pending: (reason unknown)

~ #
~ #
~ # ls
bin          initramfs.cpio.gz  root          usr
dev          linuxrc            sbin
~ #
```

Ножницы

Снимок экрана скопирован в буфер обмена
Автоматически сохраняются в папке снимков